

Miguel Blanco García



Professional Summary

PhD in Physics and materials science with an MSc in Renewable Energies and experience in technical analysis, energy technologies and international project coordination. Skilled in evaluating complex technical problems, synthesizing information from multiple sources and translating findings into clear recommendations. Background in energy storage, hydrogen technologies, fuel cells and advanced materials, combined with experience in multidisciplinary R&D collaborations.

Work History

Deutsches Elektronen-Synchrotron (DESY) - PhD Researcher

Hamburg, Germany

04/2022 - 07/2026

- Led a work package within the EU-funded CORAERO project, coordinating experimental planning, timelines, technical reporting, and collaboration with international partners.
- Managed complex experimental campaigns at international research facilities, prioritizing activities under time constraints, analyzing technical results and adapting strategies when unexpected challenges emerged.
- Translated complex experimental findings into structured reports, peer-reviewed publications (two in JACS) and presentations for multidisciplinary international audiences.

R&D Plastics Division - Intern

Madrid, Spain

09/2021 - 03/2022

- Supported industrial R&D activities within Repsol's Plastics Division, evaluating material performance, comparing formulations and translating experimental results into technical recommendations.
- Contributed to polymer formulation trials, including polypropylene grades for 3D printing, identifying anomalies and improving performance.
- Produced technical reports and testing procedures for QA.

Materials Physics Department, UAM - Research Trainee / Assistant

Madrid, Spain

09/2020 - 07/2021

- Investigated hydrogen storage in Mg-based ultrathin films, analyzing adsorption-desorption mechanisms and material stability.
- Explored photo-assisted hydrogen production using FeS₂, pyrite, and semiconductor materials, including electrochemical characterization.
- Worked on hydrogen, fuel-cell, and electrochemical materials, covering production, performance, and degradation aspects.

Languages

Spanish (native)

English (fluent)

Italian (fluent)

German (B1, improving)

Certifications

- Patentability and IP Management courses by EPO — (2025-2026)
- Management and Coordination of EU-Funded Projects with the PM2 Methodology (PM2 for EU Funds) — (2025)
- Foundations of Project Management for Scientific Projects — (2024)

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Selected Skills & Expertise

- Energy technologies and transition
- Energy storage, hydrogen and electrochemical technologies
- Technical analysis and risk identification
- Structured problem solving and data analysis
- Project and workstream coordination
- Stakeholder communication and technical reporting
- Technical literature review and technology evaluation
- Advanced materials and semiconductor technologies
- Data analysis and technical tools (Office, Python, MATLAB, Origin)
- AI-assisted workflows and automation (LLMs, Codex, n8n)

Education

07/2026

Universität Hamburg

PhD: Physics, Material Science

GPA: 10/10 Summa Cum Laude

- Materials science; light-driven reactions on semiconductor materials; international collaboration

01/2021

Universidad Autónoma De Madrid

MSc: Renewable Energies

GPA: 9.1/10 with Honours

- Broad training in renewable energy technologies, including energy systems, renewable generation, storage technologies and hydrogen.

01/2020

Universidad Autónoma De Madrid

BSc: Chemistry

GPA: 8.44/10 with Honours

- Solid foundation across chemistry; focus on materials science and organic chemistry.